Lever Test · Pupil Resources

Learning goal: I can test how the distance from the pivot changes the effort needed to lift a load.

Arrival Questions (In your book)

1. Which box needed fewer coins of effort: Box 1 or Box 2?

2. What do we call the fixed point the ruler turns on?

3. What stays the same in every trial of our test?

4. Can you ask an adult to help?

Starter Question

What pattern do the results (5 cm -> 8 coins, 10 cm -> 4 coins, 20 cm -> 2 coins) show?

A. As the effort distance from the pivot increases, the effort force needed to lift the load decreases.

B. Moving further from the pivot makes the load heavier.

C. Distance from the pivot makes no difference at all.

Method Strip

1. Place the pivot under the 10 cm mark of the ruler and check the load is fixed 5 cm from the pivot.

2. Stack identical coins 5 cm on the other side of the pivot until the load just lifts. STOP AND CHECK.

3. Record the coin count, then repeat with the effort stack at 10 cm from the pivot.

4. Repeat with the effort stack at 20 cm from the

pivot.

5. Record all three results in your book and write the pattern.

Exit Ticket

Which effort distance needed the fewest coins: 5 cm or 20 cm?

What happens to the effort needed as you push further from the pivot?

Name two things we kept the same to make the lever test fair.